2.02 Complex Numbers: Exponential Form

Worked Solutions

1 a)

$$3\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$$

1 b)

$$5\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$$

1 c)

$$2\left(\cos(-\frac{\pi}{3})+\mathrm{i}\sin(-\frac{\pi}{3})\right)=2\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)$$

1 d)

$$7\left(\cos\frac{18\pi}{5}+\mathrm{i}\sin\frac{18\pi}{5}\right)=7\left(\cos(-\frac{2\pi}{5})+\mathrm{i}\sin(-\frac{2\pi}{5})\right)$$

1 e)

$$-4\left(\cos\frac{3\pi}{4}+\mathrm{i}\sin\frac{3\pi}{4}\right)=4\left(\cos(-\frac{\pi}{4})+\mathrm{i}\sin(-\frac{\pi}{4})\right)$$

1 f)

$$-2\left(\cos\frac{5\pi}{4}+\mathrm{i}\sin\frac{5\pi}{4}\right)=2\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$$

2 a)

$$2\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)=\sqrt{2}+\sqrt{2}\mathrm{i}$$

2 b)

$$3\left(\cos\pi + \mathrm{i}\sin\pi\right) = -3$$

2 c)

$$6\left(\cos\frac{7\pi}{3} + \mathrm{i}\sin\frac{7\pi}{3}\right) = 3 + 3\sqrt{3}\mathrm{i}$$

2 d)

$$2\left(\cos(-\pi) + \mathrm{i}\sin(-\pi)\right) = -2$$

2 e)

$$-4\left(\cos\frac{5\pi}{2} + \mathrm{i}\sin\frac{5\pi}{2}\right) = -4\mathrm{i}$$

2 f)

$$-\sqrt{2}\left(\cos(-\frac{4\pi}{3}) + \mathrm{i}\sin(-\frac{4\pi}{3})\right) = \frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2}\mathrm{i}$$

3 a)

$$4\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$$

3 b)

$$5\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}$$

3 c)

$$\sqrt{2}\mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}$$

3 d)

$$\begin{aligned} 5\left(\cos(-\frac{3\pi}{2}) + \mathrm{i}\sin(-\frac{3\pi}{2})\right) &= 5\left(\cos(\frac{\pi}{2}) + \mathrm{i}\sin(\frac{\pi}{2})\right) \\ &= 5\mathrm{e}^{\frac{\pi}{2}\mathrm{i}} \end{aligned}$$

3 e)

$$\begin{aligned} 4\left(\cos\frac{\pi}{6} - \mathrm{i}\sin\frac{\pi}{6}\right) &= 4\left(\cos(-\frac{\pi}{6}) + \mathrm{i}\sin(-\frac{\pi}{6})\right) \\ &= 4\mathrm{e}^{-\frac{\pi}{6}\mathrm{i}} \end{aligned}$$

3 f)

$$\begin{aligned} \sqrt{3}\left(\cos\frac{6\pi}{7} - \mathrm{i}\sin\frac{6\pi}{7}\right) &= \sqrt{3}\left(\cos(-\frac{6\pi}{7}) + \mathrm{i}\sin(-\frac{6\pi}{7})\right) \\ &= \sqrt{3}\mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}} \end{aligned}$$

3 g)

$$\begin{aligned} \left(\cos(-\frac{\pi}{5}) - \mathrm{i}\sin(-\frac{\pi}{5})\right) &= \left(\cos(\frac{\pi}{5}) + \mathrm{i}\sin(\frac{\pi}{5})\right) \\ &= \mathrm{e}^{\frac{\pi}{5}\mathrm{i}} \end{aligned}$$

3 h)

$$\begin{aligned} 8\left(\cos(-\frac{5\pi}{4}) - \mathrm{i}\sin(-\frac{5\pi}{4})\right) &= 8\left(\cos(\frac{5\pi}{4}) + \mathrm{i}\sin(\frac{5\pi}{4})\right) \\ &= 8\left(\cos(-\frac{3\pi}{4}) + \mathrm{i}\sin(-\frac{3\pi}{4})\right) \\ &= 8\mathrm{e}^{-\frac{3\pi}{4}\mathrm{i}} \end{aligned}$$

4 a)

$|z| = \sqrt{(2)^2 + (2)^2} = 2\sqrt{2}$ and $\arg z = \arctan\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{\pi}{4}$

$$\therefore z = 2\sqrt{2}\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$$

4 b)

$|z| = \sqrt{4^2 + (-4\sqrt{3})^2} = 8$ and $\arg z = -\arctan\left(\frac{4\sqrt{3}}{4}\right) = -\frac{\pi}{3}$

$$\therefore z = 8\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}}$$

4 c)

$|z| = \sqrt{(-2)^2 + (\sqrt{2})^2} = \sqrt{6}$ and $\arg z = \pi - \arctan\left(\frac{\sqrt{2}}{2}\right) = 2.53$

$$\therefore z = \sqrt{6}\mathrm{e}^{2.53\mathrm{i}}$$

4 d)

$|z| = \sqrt{(-5)^2 + (-12)^2} = 13$ and $\arg z = -\pi + \arctan\left(\frac{12}{5}\right) = -1.97$

$$\therefore z = 13\mathrm{e}^{-1.97\mathrm{i}}$$

4 e)

$|z| = 7$ and $\arg z = \pi$

$$\therefore z = 7\mathrm{e}^{\pi\mathrm{i}}$$

4 f)

$|z| = 6$ and $\arg z = \frac{\pi}{2}$

$$\therefore z = 6\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$$

4 g)

$|z| = 8$ and $\arg z = 0$

$$\therefore z = 8\mathrm{e}^{0\mathrm{i}}$$

4 h)

$|z| = \sqrt{3^2 + 1^2} = \sqrt{10}$ and $\arg z = \arctan\left(\frac{1}{3}\right) = 0.322$

$$\therefore z = \sqrt{10}\mathrm{e}^{.322\mathrm{i}}$$

4 i)

$|z| = \sqrt{1^2 + (-\sqrt{5})^2} = \sqrt{6}$ and $\arg z = -\arctan\left(\frac{\sqrt{5}}{1}\right) = -1.15$

$$\therefore z = \sqrt{6}\mathrm{e}^{-1.15\mathrm{i}}$$

5 a)

$$z^* = 2\mathrm{e}^{-\frac{\pi}{5}\mathrm{i}}$$

5 b)

$z = 8\mathrm{e}^{-\frac{\pi}{2}\mathrm{i}}$ So $z^* = 8\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$

5 c)

$z = 3\mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}}$ So $z^* = 3\mathrm{e}^{\frac{6\pi}{7}\mathrm{i}}$

5 d)

$z = -6\mathrm{e}^{\frac{2\pi}{7}\mathrm{i}} = 6\mathrm{e}^{-\frac{5\pi}{7}\mathrm{i}}$ So $z^* = 6\mathrm{e}^{\frac{5\pi}{7}\mathrm{i}}$

6 a)

$$\begin{aligned}\mathrm{e}^{\frac{\pi}{4}\mathrm{i}+\frac{\pi}{2}\mathrm{i}} &= \mathrm{e}^{\frac{3\pi}{4}\mathrm{i}} \\ &= \left(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\right) \\ &= -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}\end{aligned}$$

6 b)

$$\begin{aligned}2\sqrt{3}\mathrm{e}^{\frac{\pi}{3}\mathrm{i}+\frac{\pi}{5}\mathrm{i}} &= 2\sqrt{3}\mathrm{e}^{\frac{8\pi}{15}\mathrm{i}} \\ &= 2\sqrt{3}\left(\cos\frac{8\pi}{15} + \mathrm{i}\sin\frac{8\pi}{15}\right) \\ &= -0.362 + 3.45\mathrm{i}\end{aligned}$$

6 c)

$$\begin{aligned}-2\mathrm{e}^{2\mathrm{i}+5\mathrm{i}} &= -2\mathrm{e}^{7\mathrm{i}} \\ &= -2\left(\cos 7 + \mathrm{i}\sin 7\right) \\ &= -1.51 - 1.31\mathrm{i}\end{aligned}$$

6 d)

$$\begin{aligned}\mathrm{e}^{\frac{\pi}{4}\mathrm{i}-\frac{\pi}{2}\mathrm{i}+\frac{\pi}{3}\mathrm{i}} &= \mathrm{e}^{\frac{\pi}{12}\mathrm{i}} \\ &= \left(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\right) \\ &= \frac{\sqrt{6}+\sqrt{2}}{4} + \left(\frac{\sqrt{6}-\sqrt{2}}{4}\right)\mathrm{i}\end{aligned}$$

6 e)

$$\begin{aligned}2\sqrt{2}\mathrm{e}^{-\frac{\pi}{6}\mathrm{i}+\frac{\pi}{3}\mathrm{i}+\frac{5\pi}{4}\mathrm{i}} &= 2\sqrt{2}\mathrm{e}^{\frac{17\pi}{12}\mathrm{i}} \\ &= 2\sqrt{2}\left(\cos\frac{17\pi}{12} + \mathrm{i}\sin\frac{17\pi}{12}\right) \\ &= 1 - \sqrt{3} - (1+\sqrt{3})\mathrm{i}\end{aligned}$$

7 a)

$$\begin{aligned} e^{\frac{\pi}{4}i-\frac{\pi}{2}i} &= e^{-\frac{\pi}{4}i} \\ &= \left(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})\right) \\ &= \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \end{aligned}$$

7 b)

$$\begin{aligned} \frac{1}{3}e^{\frac{\pi}{3}i-\frac{\pi}{4}i} &= \frac{1}{3}e^{\frac{\pi}{12}i} \\ &= \frac{1}{3}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) \\ &= \frac{\sqrt{6}+\sqrt{2}}{12} + \left(\frac{\sqrt{6}-\sqrt{2}}{12}\right)i \end{aligned}$$

7 c)

$$\begin{aligned} \frac{1}{5}e^{3i-7i} &= \frac{1}{5}e^{-4i} \\ &= \frac{1}{5}\left(\cos(-4) + i\sin(-4)\right) \\ &= -0.131 + 0.151i \end{aligned}$$

7 d)

$$\begin{aligned} 4e^{\frac{\pi}{3}i+\frac{\pi}{4}i-\frac{\pi}{2}i} &= 4e^{\frac{\pi}{12}i} \\ &= 4\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right) \\ &= \sqrt{6}+\sqrt{2}+(\sqrt{6}-\sqrt{2})i \end{aligned}$$

7 e)

$$\begin{aligned} \frac{\sqrt{2}}{3}e^{-\frac{\pi}{4}i-\frac{\pi}{5}i+\frac{4\pi}{7}i} &= \frac{\sqrt{2}}{3}e^{\frac{17\pi}{140}i} \\ &= \frac{\sqrt{2}}{3}\left(\cos\frac{17\pi}{140} + i\sin\frac{17\pi}{140}\right) \\ &= 0.438 + 0.176i \end{aligned}$$

8)

$$e^{3\theta i} \times e^{2\theta i} = e^{5\theta i}$$

9 a)

$$e^{\frac{3\pi}{7}i} \times e^{\frac{4\pi}{7}i} = e^{\pi i}$$

9 b)

$$e^{\frac{\pi}{4}i} \times e^{-\frac{\pi}{3}i} = e^{-\frac{\pi}{12}i}$$

9 c)

$$3e^{-\frac{\pi}{4}i} \times e^{\frac{3\pi}{8}i} = 3e^{\frac{\pi}{8}i}$$

9 d)

$$\begin{aligned} 4\left(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})\right)\left(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})\right) &= 4e^{-\frac{\pi}{3}i} \times e^{-\frac{\pi}{4}i} \\ &= 4e^{-\frac{7\pi}{12}i} \end{aligned}$$

10)

$$\frac{e^{6\theta i}}{e^{4\theta i}} = e^{2\theta i}$$

11 a)

$$\frac{e^{\frac{2\pi}{5}i}}{e^{\frac{4\pi}{5}i}} = e^{-\frac{2\pi}{5}i}$$

11 b)

$$\frac{e^{\frac{\pi}{4}i}}{e^{-\frac{2\pi}{3}i}} = e^{\frac{11\pi}{12}i}$$

11 c)

$$\begin{aligned}\frac{\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}}{\cos\frac{\pi}{3}-\mathrm{i}\sin\frac{\pi}{3}} &= \frac{\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}}{\cos(-\frac{\pi}{3})+\mathrm{i}\sin(-\frac{\pi}{3})}\\ &= \frac{\mathrm{e}^{\frac{\pi}{5}\mathrm{i}}}{\mathrm{e}^{-\frac{\pi}{3}\mathrm{i}}}\\ &= \mathrm{e}^{\frac{8\pi}{15}\mathrm{i}}\end{aligned}$$

12 a)

$$|z| = \sqrt{2^2+(-2)^2} = 2\sqrt{2}$$

$$\arg z = -\arctan\left(\frac{2}{2}\right) = -\frac{\pi}{4}$$

$$\therefore z = 2\sqrt{2}\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}$$

12 b)

$$|w| = \sqrt{1^2+(\sqrt{3})^2} = 2$$

$$\arg w = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$$

$$\therefore w = 2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$$

12 c)

$$\begin{aligned}z^2+w^2 = (2\sqrt{2}\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}})^2+(2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}})^2 &= 8\mathrm{e}^{-\frac{\pi}{2}\mathrm{i}}+4\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}}\\ &= 8\left(\cos(-\frac{\pi}{2})+\mathrm{i}\sin(-\frac{\pi}{2})\right)+4\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)\end{aligned}$$

So that

$$a = 8\cos\left(-\frac{\pi}{2}\right)+4\cos\frac{2\pi}{3} = -2$$

and

$$b = 8\sin\left(-\frac{\pi}{2}\right)+4\sin\frac{2\pi}{3} = -8+2\sqrt{3}$$

Which gives:

$$\frac{b}{a} = \frac{-8+2\sqrt{3}}{-2} = 4-\sqrt{3} \quad \text{as req.}$$

13 a)

$$3\sqrt{2}$$

13 b)

$$\frac{3}{\sqrt{2}} \text{ or } \frac{3\sqrt{2}}{2}$$

13 c)

$$\frac{\pi}{4} + \frac{\pi}{3} = \frac{7\pi}{12}$$

13 d)

$$\frac{\pi}{4} - \frac{\pi}{3} = -\frac{\pi}{12}$$

14 a)

$$3\sqrt{3}$$

14 b)

$$\frac{\sqrt{3}}{3}$$

14 c)

$$-\frac{2\pi}{3} + \frac{\pi}{5} = -\frac{7\pi}{15}$$

14 d)

$$-\frac{2\pi}{3} - \frac{\pi}{5} = -\frac{13\pi}{15}$$

15 a)

$$z_1 z_2 = r_1 \mathrm{e}^{\theta_1 \mathrm{i}} \times r_2 \mathrm{e}^{\theta_2 \mathrm{i}} = r_1 r_2 \mathrm{e}^{(\theta_1 + \theta_2)\mathrm{i}}$$

So $|z_1 z_2| = r_1 r_2 = |z_1||z_2|$ as req.

15 b)

$$\frac{z_1}{z_2} = \frac{r_1 \mathrm{e}^{\theta_1 \mathrm{i}}}{r_2 \mathrm{e}^{\theta_2 \mathrm{i}}} = \frac{r_1}{r_2} \mathrm{e}^{(\theta_1 - \theta_2)\mathrm{i}}$$

So $\left|\frac{z_1}{z_2}\right| = \frac{r_1}{r_2} = \frac{|z_1|}{|z_2|}$ as req.

15 c)

$$z_1 z_2 = r_1 \mathrm{e}^{\theta_1 \mathrm{i}} \times r_2 \mathrm{e}^{\theta_2 \mathrm{i}} = r_1 r_2 \mathrm{e}^{(\theta_1+\theta_2)\mathrm{i}}$$

So $\arg(z_1 z_2) = \theta_1 + \theta_2 = \arg(z_1) + \arg(z_2)$ as req.

15 d)

$$\frac{z_1}{z_2} = \frac{r_1 \mathrm{e}^{\theta_1 \mathrm{i}}}{r_2 \mathrm{e}^{\theta_2 \mathrm{i}}} = \frac{r_1}{r_2} \mathrm{e}^{(\theta_1-\theta_2)\mathrm{i}}$$

So $\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2 = \arg(z_1) - \arg(z_2)$ as req.

16 a)

$$|z| = \sqrt{(-6)^2 + (6\sqrt{3})^2} = 12$$

$$\arg z = \pi - \arctan\left(\frac{6\sqrt{3}}{6}\right) = \frac{2\pi}{3}$$

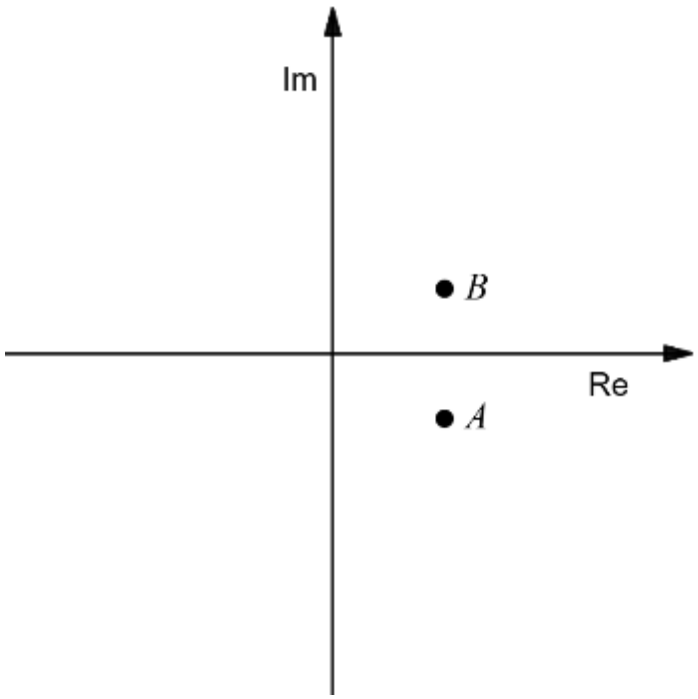
So $z = 12\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}}$ and $w = 4\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}}$

$$zw = 12\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} \times 4\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}} = 48\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$$

16 b)

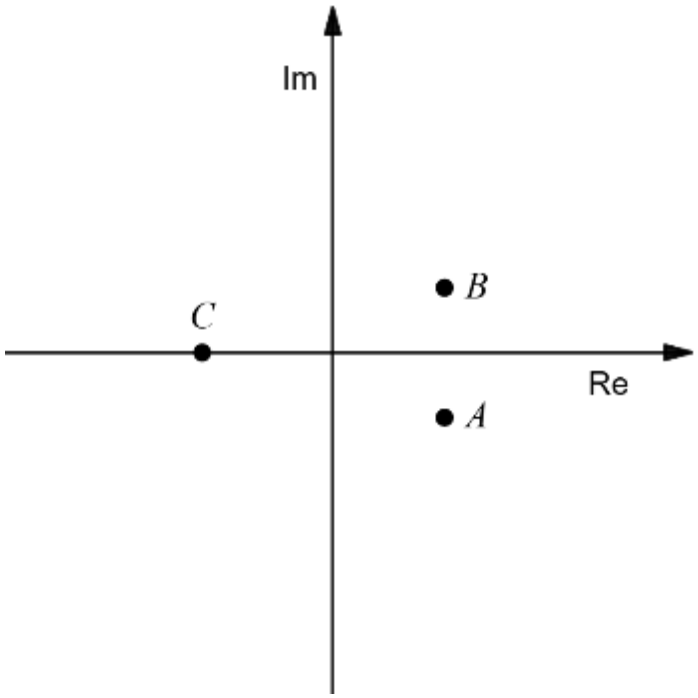
$$\begin{aligned} \frac{z}{w} &= \frac{12\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}}}{4\mathrm{e}^{-\frac{5\pi}{12}\mathrm{i}}} \\ &= 3\mathrm{e}^{\frac{13\pi}{12}\mathrm{i}} \\ &= 3\mathrm{e}^{-\frac{11\pi}{12}\mathrm{i}} \end{aligned}$$

17 a)

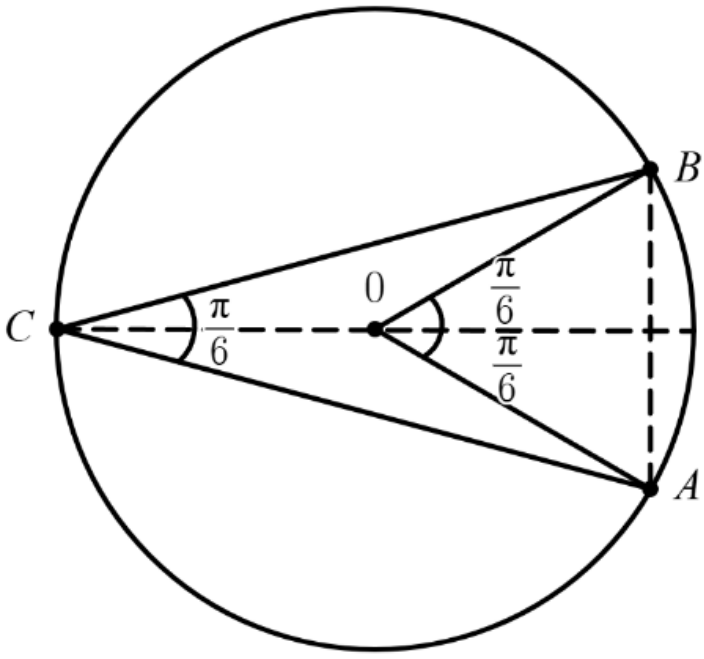


17 b)

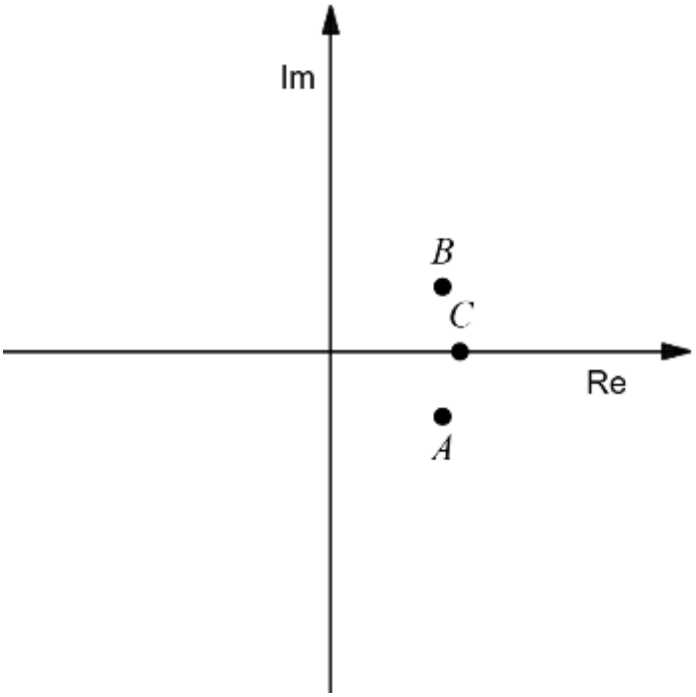
A is given by $4\left(\cos(-\frac{\pi}{6}) + \mathrm{i}\sin(-\frac{\pi}{6})\right) = 2\sqrt{3} - 2\mathrm{i}$ B is its complex conjugate: $2\sqrt{3} + 2\mathrm{i}$



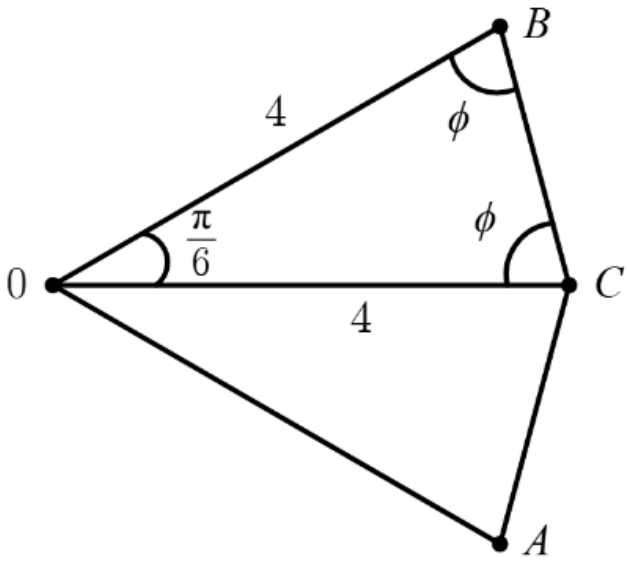
Since all of the points have a modulus of 4, they all lie on a circle: If C is -4 then



And so $\angle ACB = 2 \times \frac{\pi}{12} = \frac{\pi}{6}$



If C is 4 then



And so $\angle ACB = \frac{1}{2}\left(\pi - \frac{\pi}{6}\right) \times 2 = \frac{5\pi}{6}$ So the possible angles are $\frac{\pi}{6}$ or $\frac{5\pi}{6}$

18 a)

$$\begin{aligned}\frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}) &= \frac{1}{2}\mathrm{e}^{\mathrm{i}\theta} + \frac{1}{2}\mathrm{e}^{-\mathrm{i}\theta} \\ &= \frac{1}{2}(\cos\theta + \mathrm{i}\sin\theta) + \frac{1}{2}(\cos\theta - \mathrm{i}\sin\theta) \\ &= \frac{1}{2}\cos\theta + \frac{1}{2}\mathrm{i}\sin\theta + \frac{1}{2}\cos\theta - \frac{1}{2}\mathrm{i}\sin\theta \\ &= \cos\theta \quad \text{as req.}\end{aligned}$$

18 b)

$$\begin{aligned}\frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})\times\frac{\mathrm{i}}{\mathrm{i}} &= -\frac{\mathrm{i}}{2}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})\\ &= -\frac{\mathrm{i}}{2}\mathrm{e}^{\mathrm{i}\theta}+\frac{\mathrm{i}}{2}\mathrm{e}^{-\mathrm{i}\theta}\\ &= -\frac{\mathrm{i}}{2}(\cos\theta+\mathrm{i}\sin\theta)+\frac{\mathrm{i}}{2}(\cos\theta-\mathrm{i}\sin\theta)\\ &= -\frac{\mathrm{i}}{2}\cos\theta-\frac{\mathrm{i}^2}{2}\sin\theta+\frac{\mathrm{i}}{2}\cos\theta-\frac{\mathrm{i}^2}{2}\sin\theta\\ &= \frac{1}{2}\sin\theta+\frac{1}{2}\sin\theta\\ &= \sin\theta \quad \text{as req.}\end{aligned}$$

Alternatively:

$$\begin{aligned}\mathrm{e}^{\mathrm{i}\theta} &= \cos\theta+\mathrm{i}\sin\theta \qquad (1)\\ \mathrm{e}^{-\mathrm{i}\theta} &= \cos(-\theta)+\mathrm{i}\sin(-\theta)\\ &= \cos\theta-\mathrm{i}\sin\theta \qquad (2)\end{aligned}$$

(1) + (2) $\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}=2\cos\theta$

$$\Rightarrow \cos\theta=\frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})$$

(1) − (2) $\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta}=2\mathrm{i}\sin\theta$

$$\Rightarrow \sin\theta=\frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})$$

19 a)

$\mathrm{e}^{\mathrm{i}\theta}=\cos\theta+\mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta}=\cos\theta-\mathrm{i}\sin\theta$

Multiply the LHS:

$$\mathrm{e}^{\mathrm{i}\theta}\times\mathrm{e}^{-\mathrm{i}\theta}=1$$

Multiply the RHS:

$$\begin{aligned}(\cos\theta+\mathrm{i}\sin\theta)(\cos\theta-\mathrm{i}\sin\theta) &= \cos^2\theta+\mathrm{i}\sin\theta\cos\theta-\mathrm{i}\sin\theta\cos\theta+\sin^2\theta\\ &= \cos^2\theta+\sin^2\theta\end{aligned}$$

$$\therefore \sin^2\theta+\cos^2\theta=1 \quad \text{as req.}$$

19 b)

Since $\mathrm{e}^{\mathrm{i}\theta}=\cos\theta+\mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta}=\cos\theta-\mathrm{i}\sin\theta$

Then $\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}=2\cos\theta$ so $\cos\theta=\frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})$

and $\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta}=2\mathrm{i}\sin\theta$ so $\sin\theta=\frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})$

So that:

$$\begin{aligned}2\sin\theta\cos\theta &= 2\times\frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})\times\frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})\\ &= \frac{1}{2\mathrm{i}}(\mathrm{e}^{2\mathrm{i}\theta}-\mathrm{e}^{-2\mathrm{i}\theta})\\ &= \sin 2\theta \quad \text{as req.}\end{aligned}$$

19 c)

Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

Then $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ so $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$

and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$ so $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$

So that:

$$\begin{aligned}\cos^2\theta - \sin^2\theta &= \frac{1}{4}(e^{i\theta} + e^{-i\theta})^2 + \frac{1}{4}(e^{i\theta} - e^{-i\theta})^2 \\ &= \frac{1}{4}(e^{2i\theta} + 2 + e^{-2i\theta}) + \frac{1}{4}(e^{2i\theta} - 2 + e^{-2i\theta}) \\ &= \frac{1}{2}(e^{2i\theta} + e^{-2i\theta}) \\ &= \cos 2\theta \quad \text{as req.}\end{aligned}$$

20)

$$(\cos\theta + i\sin\theta)^n = (e^{i\theta})^n = e^{in\theta} = \cos n\theta + i\sin n\theta$$

21)

$$\begin{aligned}e^{z} &= e^{a+bi} = e^{a}e^{bi} = e^{a}(\cos b + i\sin b) \\ e^{z^*} &= e^{a-bi} = e^{a}e^{-bi} = e^{a}(\cos b - i\sin b) \\ (e^{z})^* &= e^{a}(\cos b - i\sin b)\end{aligned}$$

$$\therefore e^{z^*} = (e^{z})^* \quad \text{as req.}$$

Alternatively:

$$\begin{aligned}e^{z^*} &= e^{a-bi} = e^{a}e^{-bi} \\ &= e^{a}(\cos(-b) + i\sin(-b)) \\ &= e^{a}(\cos b - i\sin b) \\ &= e^{a}(e^{bi})^* = (e^{a+bi})^* = (e^{z})^*\end{aligned}$$

22)

$e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

Factor out $e^{i\theta}$:

$$e^{2i\theta} - 1 = e^{i\theta}(e^{i\theta} - e^{-i\theta})$$

Since $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$:

$$\begin{aligned}e^{2i\theta} - 1 &= e^{i\theta}(2i\sin\theta) \\ &= 2ie^{i\theta}\sin\theta \quad \text{as req.}\end{aligned}$$

23)

$$\begin{aligned}\left(1-\frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}\right)\left(1-\frac{1}{4}\mathrm{e}^{-\mathrm{i}\theta}\right) &= 1-\frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}-\frac{1}{4}\mathrm{e}^{-\mathrm{i}\theta}+\frac{1}{16}\\ &= \frac{17}{16}-\frac{1}{2}\times\frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})\\ &= \frac{17}{16}-\frac{1}{2}\cos\theta\end{aligned}$$

24)

$$\begin{aligned}(1-a\mathrm{e}^{\mathrm{i}\theta})(1-a\mathrm{e}^{-\mathrm{i}\theta}) &= 1-a\mathrm{e}^{\mathrm{i}\theta}-a\mathrm{e}^{-\mathrm{i}\theta}+a^2\\ &= 1+a^2-2a\times\frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})\\ &= 1+a^2-2a\cos\theta\end{aligned}$$

25)

$$\begin{aligned}z+w &= 2\mathrm{e}^{\frac{\pi}{6}\mathrm{i}}+2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}\\ &= 2\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)+2\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)\\ &= 1+\sqrt{3}+(1+\sqrt{3})\mathrm{i}\\ |z| &= \sqrt{(1+\sqrt{3})^2+(1+\sqrt{3})^2}=\sqrt{2(1+\sqrt{3})^2}=\sqrt{2}(1+\sqrt{3})\\ \arg z &= \arctan\left(\frac{1+\sqrt{3}}{1+\sqrt{3}}\right)=\frac{\pi}{4}\\ \therefore z &= \sqrt{2}(1+\sqrt{3})\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}=(\sqrt{2}+\sqrt{6})\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}\quad\text{as req.}\end{aligned}$$